

Set Theory for Computer Science

Week 5: Functions and Cardinality

Functions

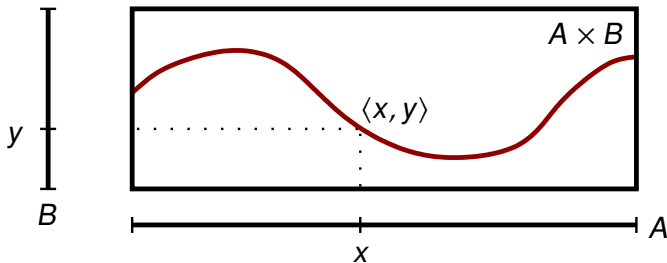
Composition and inverse

Counting and cardinality

Countable and uncountable sets

Functions

A **function** $f: A \rightarrow B$ is a binary relation of type $A \times B$ such that every $x \in A$ relates to at most one $y \in B$



Notation: $f(x) = y$ instead of $\langle x, y \rangle \in f$

Domain, codomain and range

A **function** $f: A \rightarrow B$ is a binary relation of type $A \times B$ such that every $x \in A$ relates to at most one $y \in B$

domain A

codomain B

domain of definition $D_f := \{x \in A : f(x) = y \text{ for some } y \in B\}$

range (or image) $R_f := \{y \in B : y = f(x) \text{ for some } x \in A\}$

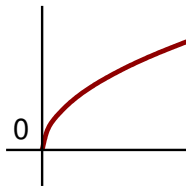
- ▶ if $D_f = A$ then f is called **total**
- ▶ if $R_f = B$ then f is called **surjective**

Domain, codomain and range

A **function** $f: A \rightarrow B$ is a binary relation of type $A \times B$ such that every $x \in A$ relates to at most one $y \in B$

Examples

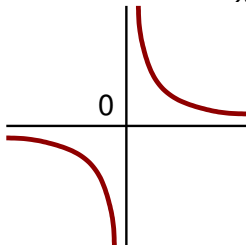
$f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \sqrt{x}$



$$D_f = \{x: x \geq 0\}$$

$$R_f = \{y: y \geq 0\}$$

$g: \mathbb{R} \rightarrow \mathbb{R}$ with $g(x) = \frac{1}{x}$



$$D_g = \mathbb{R} \setminus \{0\}$$

$$R_g = \mathbb{R} \setminus \{0\}$$

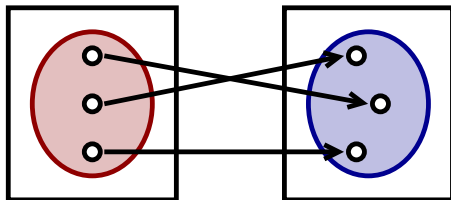
Injections and surjections

$f: A \rightarrow B$ is **injective** if

$$x_1 \neq x_2 \rightarrow f(x_1) \neq f(x_2)$$

or equivalently, if

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2$$

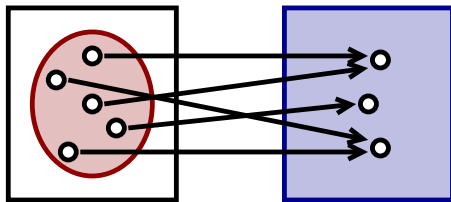


A

B

$f: A \rightarrow B$ is **surjective** if

$$R_f = B$$



Injections and surjections: example

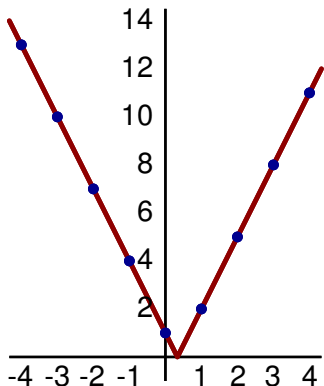
Note: choice of domain/codomain matters!

$f: \mathbb{R} \rightarrow \{y: y \geq 0\}$ with $f(x) = |3x - 1|$

- ▶ not injective
- ▶ surjective

$g: \mathbb{Z} \rightarrow \mathbb{N}$ with $g(x) = |3x - 1|$

- ▶ injective
- ▶ not surjective



Exercise

Is the given function injective? surjective? Explain!

(a) $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = e^x$

(b) $g: \mathbb{N}^2 \rightarrow \mathbb{Z}$ with $g(\langle n, m \rangle) = n - m$

Functions

Composition and inverse

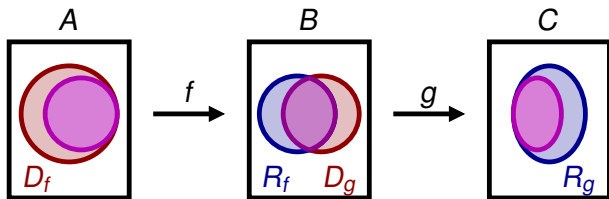
Counting and cardinality

Countable and uncountable sets

Composition of functions

The **composition** $g \circ f$ of $f: A \rightarrow B$ and $g: B \rightarrow C$ satisfies

$$f(x) = y \wedge g(y) = z \quad \Rightarrow \quad (g \circ f)(x) = g(f(x)) = z$$



Note: $D_{g \circ f} \subseteq D_f$ and it is possible that $D_{g \circ f} \neq D_f$
 $R_{g \circ f} \subseteq R_g$ and it is possible that $R_{g \circ f} \neq R_g$

Composition of functions: example

Example

Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 1 - x^2$$

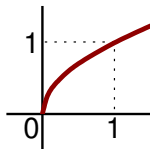
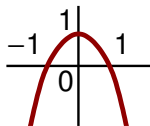
$$D_f = \mathbb{R}$$

$$R_f = \{y: y \leq 1\}$$

$$g(x) = \sqrt{x}$$

$$D_g = \{x: x \geq 0\}$$

$$R_g = \{y: y \geq 0\}$$



Composition of functions: example

Example

Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 1 - x^2$$

$$D_f = \mathbb{R}$$

$$R_f = \{y: y \leq 1\}$$

$$g(x) = \sqrt{x}$$

$$D_g = \{x: x \geq 0\}$$

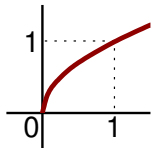
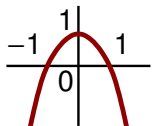
$$R_g = \{y: y \geq 0\}$$

Composition

$$(g \circ f)(x) = g(f(x)) = \sqrt{1 - x^2}$$

$$D_{g \circ f} = \{x: -1 \leq x \leq 1\} \neq D_f$$

$$R_{g \circ f} = \{y: 0 \leq y \leq 1\} \neq R_g$$

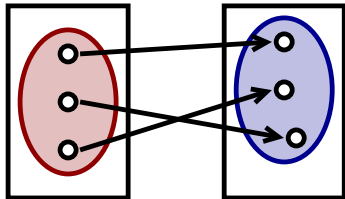


Inverse functions

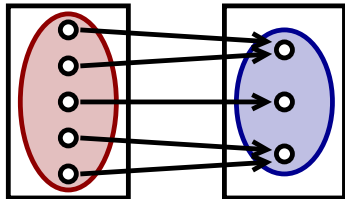
- ▶ $f: A \rightarrow B$ considered as a *relation* has an inverse
- ▶ this inverse *relation* is not necessarily a *function*

Theorem

The inverse of f is a function if and only if f is injective



injective: **one-one** or **1-1**



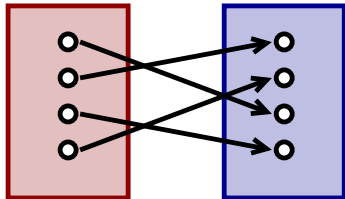
non-injective: **many-one**

Bijections

- ▶ a *total* injection $f: A \rightarrow B$ has an inverse function $f^{-1}: B \rightarrow A$
- ▶ this inverse function is not necessarily *total*

Theorem

The inverse of a total injection f is total if and only if f is surjective



$f: A \rightarrow B$ is **bijective** if

- ▶ f is total
- ▶ f is injective
- ▶ f is surjective

Bijections have a bijective inverse

Exercise

Consider the following functions of type $\mathbb{R} \rightarrow \mathbb{R}$:

$$\text{sqr}(x) = x^2$$

$$\text{double}(x) = 2x$$

$$\text{dec}(x) = x - 1$$

$$\text{inv}(x) = 1/x$$

- (a) Give a description (an expression for $f(x)$) of the function $f := \text{inv} \circ \text{dec}^{-1} \circ \text{sqr}$ and specify D_f and R_f
- (b) Express $g: x \mapsto \frac{1}{2}x^2 - 1$ as an explicit composition of the functions above and their inverses

break

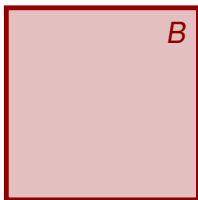
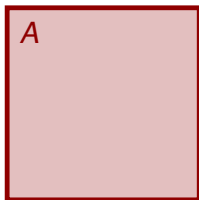
Functions

Composition and inverse

Counting and cardinality

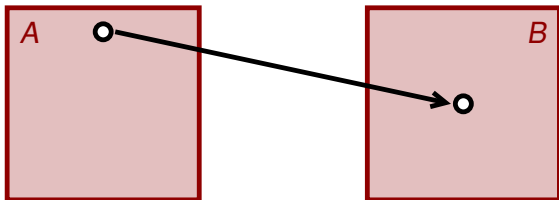
Countable and uncountable sets

How to compare the sizes of two sets?



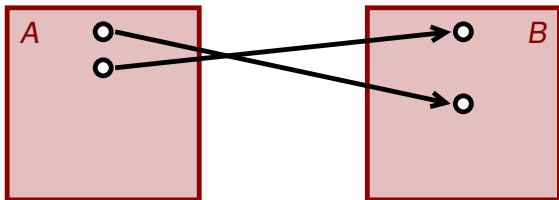
Idea: try to map *every* element of A to a *distinct* element of B

How to compare the sizes of two sets?



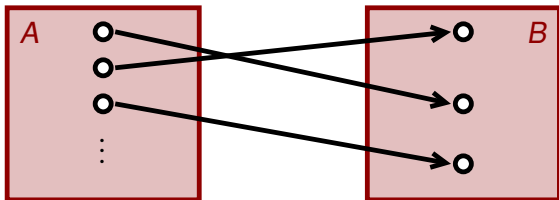
Idea: try to map *every* element of A to a *distinct* element of B

How to compare the sizes of two sets?



Idea: try to map *every* element of A to a *distinct* element of B

How to compare the sizes of two sets?



Idea: try to map *every* element of A to a *distinct* element of B

- ▶ **success** $\Rightarrow$ A cannot have more elements than B
- ▶ **failure** $\Rightarrow$ A must have more elements than B

The concept of cardinality

The **cardinality** $|A|$ of A is an abstract notion of the size of a set A

Definition of cardinality

$|A| \leq |B| \iff$ there exists a total injection $f: A \rightarrow B$

$|A| = |B| \iff$ there exists a bijection $f: A \rightarrow B$

Special case: finite sets

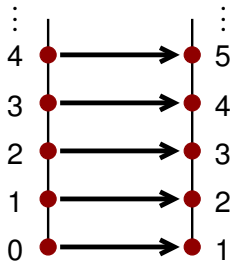
$|A| = \#A = m \iff$ there exists a bijection $f: \{1, \dots, m\} \rightarrow A$

Examples: sets with the cardinality of $\mathbb{N}$

Examples

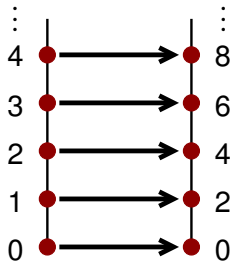
$$A := \{1, 2, 3, 4, \dots\} = \mathbb{N} \setminus \{0\}$$

$$\text{bijection } \mathbb{N} \rightarrow A: f(n) = n + 1$$



$$B := \{0, 2, 4, 6, \dots\} = 2\mathbb{N}$$

$$\text{bijection } \mathbb{N} \rightarrow B: g(n) = 2n$$



Conclusion: $|\mathbb{N}| = |\mathbb{N} \setminus \{0\}| = |2\mathbb{N}|$

The Cantor–Schröder–Bernstein theorem

Theorem

If there exist total injective functions $f: A \rightarrow B$ and $g: B \rightarrow A$ then there exists a bijection $h: A \rightarrow B$

Hence: $|A| \leq |B| \wedge |B| \leq |A| \rightarrow |A| = |B|$ (anti-symmetry)

The Cantor–Schröder–Bernstein theorem

Theorem

If there exist total injective functions $f: A \rightarrow B$ and $g: B \rightarrow A$ then there exists a bijection $h: A \rightarrow B$

Hence: $|A| \leq |B| \wedge |B| \leq |A| \rightarrow |A| = |B|$ (anti-symmetry)

Also: $|A| \leq |A|$ (reflexivity)

$|A| \leq |B| \wedge |B| \leq |C| \rightarrow |A| \leq |C|$ (transitivity)

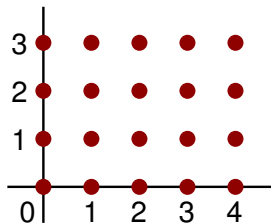
Conclusion: $\leq$ is a partial order on cardinalities of sets

$\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$ have the same cardinality

Claim

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$

Proof



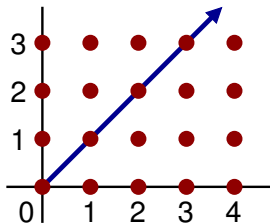
$\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$ have the same cardinality

Claim

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$

Proof

- total injection $f: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}: f(n) = \langle n, n \rangle$



$\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$ have the same cardinality

Claim

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$

Proof

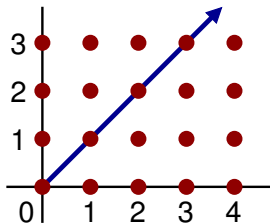
► total injection $f: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}: f(n) = \langle n, n \rangle$

► total injection $g: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}: g(\langle n, k \rangle) = 2^n 3^k$

proof of injectivity: $2^{n_1} 3^{k_1} = 2^{n_2} 3^{k_2} \quad (n_2 \geq n_1)$

$$\Rightarrow 3^{k_1} = 2^{n_2 - n_1} 3^{k_2} \quad (\text{must be odd!})$$

$$\Rightarrow n_1 = n_2 \wedge k_1 = k_2$$

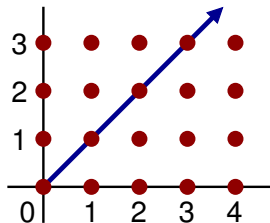


$\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$ have the same cardinality

Claim

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$

Proof



► total injection $f: \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}: f(n) = \langle n, n \rangle$

► total injection $g: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}: g(\langle n, k \rangle) = 2^n 3^k$

proof of injectivity: $2^{n_1} 3^{k_1} = 2^{n_2} 3^{k_2} \quad (n_2 \geq n_1)$

$$\Rightarrow 3^{k_1} = 2^{n_2 - n_1} 3^{k_2} \quad (\text{must be odd!})$$

$$\Rightarrow n_1 = n_2 \wedge k_1 = k_2$$

► $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ by Cantor–Schröder–Bernstein

□

Functions

Composition and inverse

Counting and cardinality

Countable and uncountable sets

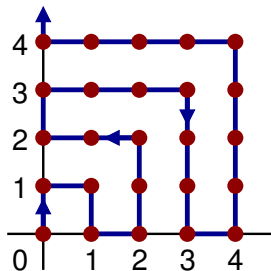
A set A is called

- ▶ **countably infinite** if there exists a bijection $f: \mathbb{N} \rightarrow A$
- ▶ **countable** if A is finite or countably infinite
- ▶ **uncountable** if A is not countable

A countable set can be *enumerated* explicitly:

$$A = \{ a_1, a_2, a_3, \dots \}$$

$\mathbb{N} \times \mathbb{N}$ is countably infinite



Enumeration of $\mathbb{N} \times \mathbb{N}$:

$\{\langle 0,0 \rangle, \langle 0,1 \rangle, \langle 1,1 \rangle, \langle 1,0 \rangle, \langle 2,0 \rangle, \langle 2,1 \rangle, \langle 2,2 \rangle, \langle 1,2 \rangle, \dots\}$

Uncountability of $(0, 1)$

Theorem: The set $(0, 1) := \{x : 0 < x < 1\}$ cannot be enumerated

Proof: Suppose $\{x_1, x_2, x_3, \dots\}$ is an enumeration of $(0, 1)$:

x_1	0	.	3	1	4	1	5	...
x_2	0	.	2	7	1	8	2	...
x_3	0	.	9	8	6	9	6	...
x_4	0	.	7	3	8	1	0	...
x_5	0	.	3	1	0	0	6	...
$\vdots$	$\vdots$							$\ddots$

Uncountability of $(0, 1)$

Theorem: The set $(0, 1) := \{x : 0 < x < 1\}$ cannot be enumerated

Proof: Suppose $\{x_1, x_2, x_3, \dots\}$ is an enumeration of $(0, 1)$:

x_1	0 .	3	1	4	1	5	...	construct x using rule
x_2	0 .	2	7	1	8	2	...	• decimal < 5 : $+ 2$
x_3	0 .	9	8	6	9	6	...	• decimal ≥ 5 : $- 2$
x_4	0 .	7	3	8	1	0	...	
x_5	0 .	3	1	0	0	6	...	
$\vdots$	$\vdots$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\ddots$	
x	0 .	5	5	4	3	4	...	this x is not in the list! in fact $ x - x_n > 10^{-n}$